

PRODUCT OVERVIEW

ReliabilityX360 is a decision-support application for critical plant assets. The current problem isn't lack of alerts — it's alerts that fire too late to prevent a trip, or too often to trust. Reliability360 combines domain and ML agents configured on the asset's own build with its historian and DCS tag data, so one framework covers a complete asset without a custom model rebuild for each — catching what's developing early enough to act on and specific enough to trust. Output reaches the reliability engineer as a named failure mode, an estimated time-to-failure and a prescribed action, reported against the parameter that caused it.



01 Asset Configuration

Shown: the guided compressor build, then the train diagram it generates.

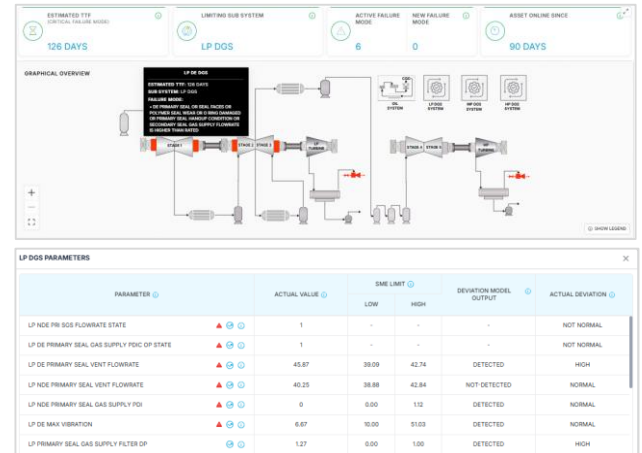
The screenshot shows the 'Machine Config' step of the guided compressor build. It includes a 'COMPRESSOR CLASSIFICATION' section with options for 'Cracked Gas Compressor', 'Ethylene Refrigeration', and 'Propylene Refrigeration'. Below this is a 'Driver Type' section with 'Steam Turbine' and 'Other' options. A 'COMPRESSOR TRAIN DIAGRAM' is shown at the bottom, illustrating the sequence of components: Turbine 1, LP Stage 1, Intermediate, HP Stage 1, Intermediate, Turbine 2, and HP Stage 2.

- Asset build** compressor service, driver type, casing count and stage grouping
- Sub-systems** dry gas seals, mechanical, oil system, turbine, condenser, process stages
- Sensor mapping** available tags mapped against a predefined list — adapts automatically
- Model depth** deviation detection, failure mode identification, failure prediction



02 Plant Data and Model Execution

Shown: the machine as a live schematic, and parameter behavior



- Inputs** Sensor tag entry, OEM/rated/design data, configuration details
- Processing** tag validation, failure mode matrix build and scoring, failure prediction
- Outputs** active failure mode, parameter forecast, estimated TTF, prescription, alerts



03 AI-Powered Model Agent Suite

Model selection is shared across assets — each model's behavior is tuned to the asset's own failure modes

- Smart Feature Engineering** Inferred tags engineered per asset from asset specific domain knowledge
- Tacit Knowledge, Encoded** Engineers' tacit knowledge built directly into every model and limit
- Deviation from Design** Tracks how equipment behavior drifts from design and rated conditions
- Early Deviation Detection** Deep learning models learns normal behavior and catches deviations early
- Failure Mode Matrix** One setup run builds the asset's symptom, mode and parameter matrix
- Failure Mode Identification** Correlates parameters against domain logic to name the failure mode
- Time-to-Failure Estimation** Estimates time to failure before a parameter crosses a threshold
- Parameter Contribution** Ranks each parameter's contribution to the active failure mode
- Sub-System Health Ranking** Scores every sub-system and names the one limiting the asset
- Workflow Suggestions** Prescribed action per mode, routed to the engineer for closeout



04 Opportunity and Operator Actions

SUB SYSTEM	ACTIVE FAILURE SYMPTOM	ACTIVE FAILURE MODE	ESTIMATED TTF	RECOMMENDATION
LP DGS	DE PRIMARY VENT FLOWRATE HIGH	DE PRIMARY SEAL OR SEAL FACES OR POLYMER SEAL WEAR OR O-RING DAMAGED OR PRIMARY SEAL HANDUP CONDITION OR SECONDARY SEAL GAS SUPPLY FLOWRATE IS HIGHER THAN RATED	126 DAYS (20-DEC-2026)	MAKE A FIELD VISIT AND CONFIRM IF THE SECONDARY SEAL GAS SUPPLY FLOWRATE IS AS PER THE RATED VALUE. IF IT IS NORMAL, THEN DETERMINE THE...
LP DGS	NDE HIGH PRIMARY SEAL GAS SUPPLY FLOW RATE	NDE PROCESS GAS LABYRINTH EXCESSIVE CLEARANCE ACTIVE SINCE: 18-APR-26	365 DAYS (28-AUG-2027)	INITIATE CALIBRATION OF PRIMARY SEAL GAS SUPPLY FLOWMETER IF IT IS CORRECT. CONTINUE OPERATIONS AS LONG AS THE REQUIRED SEAL GAS...
LP DGS	DE HIGH PRIMARY SEAL GAS SUPPLY FLOW RATE	DE PROCESS GAS LABYRINTH EXCESSIVE CLEARANCE ACTIVE SINCE: 24-APR-26	354 DAYS (19-AUG-2027)	INITIATE CALIBRATION OF PRIMARY SEAL GAS SUPPLY FLOWMETER IF IT IS CORRECT. CONTINUE OPERATIONS AS LONG AS THE REQUIRED SEAL GAS...
	NDE PRIMARY SEAL VENT	NDE SECONDARY SEAL OR SEAL FACES OR POLYMER SEAL WEAR OR O-RING BETWEEN SECONDARY SEAL GAS SUPPLY AND SECONDARY SEAL DAMAGED AND SECONDARY SEAL HANDUP CONDITION OR...		MAKE A FIELD VISIT AND CONFIRM IF THE SECONDARY SEAL GAS SUPPLY FLOWRATE...

FAILURE SYMPTOM	FAILURE MODE	PARAMETER	PARAMETER FAILURE BEHAVIOUR	SME LIMIT	CONTRIBUTION (%)
DE PRIMARY VENT FLOWRATE HIGH	DE PRIMARY SEAL OR SEAL FACES OR POLYMER SEAL WEAR OR O-RING DAMAGED OR PRIMARY SEAL HANDUP CONDITION OR SECONDARY SEAL GAS SUPPLY FLOWRATE IS HIGHER THAN RATED	LP DE PRIMARY SEAL VENT FLOWRATE	HIGH	38.09	42.74
		LP DE MAX VIBRATION	HIGH	10.00	51.03
					0

RECOMMENDATION

- MAKE A FIELD VISIT AND CONFIRM IF THE SECONDARY SEAL GAS SUPPLY FLOWRATE IS AS PER THE RATED VALUE
- IF IT IS NORMAL, THEN DETERMINE THE ACTUAL PRIMARY SEAL LEAK RATES BY SUBTRACTING THE SECONDARY SEAL FLOW RATE (PFD-RATES) FROM THE COMPENSATED PRIMARY SEAL VENT FLOW RATE
- CONFIRM SECONDARY SEAL IS IN HEALTHY CONDITION
- IF YES, PREPARE FOR A CONTROLLED SHUTDOWN AND A REPLACEMENT OF THE OSE CARTRIDGE
- IF SECONDARY SEAL IS NOT HEALTHY, THEN AN IMMEDIATE SHUTDOWN IS RECOMMENDED
- IF SEAL HANDUP IS SUSPECTED, DISCUSS WITH OEM FOR A PROCEDURE TO PRESSURIZE/DEPRESSURIZE THE SEAL CHAMBER TO RELEASE THE HANDUP
- REPLACE IT IF THIS CAN BE DONE WITH ACCEPTABLE RISK
- PLAN FOR SEAL REPLACEMENT IN AN OPPORTUNITY

30+ days

Advance warning
on every alert, before failure

\$2M+/yr

Savings potential
per critical asset protected

<2%

False alert rate
alerts an engineer can act on

Basis: avoided unplanned trips, controlled versus forced shutdown, lead time and false alert rate measured verified against actual site data



05 Platform

Historian connectivity

plug and play with leading plant historians

API Integration

enriches models with CMMS/EAM history

Daily reporting

downloadable daily reports with interactive trends

Gen AI assistant

answers plant questions in the operating workflow

Build your own dashboard

engineers assemble the views their reliability team needs

DELIVERY

Configured, validated and commissioned by Ingenero reliability engineers.

FLEXIBILITY

Any asset build; selectable model depth; selectable sensor set

SCALE

Repeat assets and sites reuse the configuration, with no/minimal rework.

Contact for a free trial:
marketing@ingenero.com